Akoizo Original Practice - 2027 Rotation Form 2 - Answer key

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A station model pressure code of 984 means:

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. 1008.4 mb

B. 998.4 mb

C. 1098.4 mb

D. 984 mb

Correct: B

2. (2 points) Which air mass is warm and humid?

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Continental tropical

B. Continental arctic

C. Maritime tropical

D. Continental polar

Correct: C

3. (2 points) Damage spreading outward in a starburst pattern indicates:

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. A tornado

B. Hail

C. A dryline

D. A downburst

Correct: D

4. (2 points) Sleet forms when:

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Melted snow refreezes in a deep cold layer near the ground

B. Snow never melts

C. Rain freezes on contact

D. Hail falls from a supercell

Correct: A

5. (2 points) In a METAR, “31/22” means:

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Pressure: Altimeter 29.72 inHg and falling quickly.

A. Temperature 31 °C, dew point 22 °C

B. Wind 310° at 22 knots

C. Visibility 31 miles

D. Pressure 31.22 inches

Correct: A

6. (2 points) A dryline separates air masses that differ mainly in:

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Wind speed only

B. Moisture

C. Temperature

D. Pressure only

Correct: B

7. (2 points) The EF scale estimates wind speed from:

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. Radar reflectivity

B. Funnel width

C. Observed damage to indicators

D. Direct anemometer readings in the tornado

Correct: C

8. (2 points) A tornado warning means:

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Conditions may become favorable later

B. The storm has ended

C. Only hail is expected

D. A tornado is occurring or imminent; take shelter now

Correct: D

9. (2 points) A long hodograph that curves clockwise near the ground indicates:

Unit: Reading weather data: station models, METARs, hodographs, and models.
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Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. No wind

B. Stable air

C. Calm conditions favoring fog

D. Strong shear favoring supercells

Correct: D

10. (2 points) Why do hurricanes weaken after landfall?

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. They lose their warm-water energy source

B. Land air is too moist

C. The jet stream stops them

D. Coriolis force becomes zero

Correct: A

11. (2 points) Interpret the adjacent opposite radial velocities cautiously.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

answer: A tight adjacent inbound/outbound couplet is evidence consistent with rotation; geometry, resolution and quality control still matter. (2 points).

12. (2 points) What additional interpretation is supported by low correlation coefficient collocated with the velocity couplet?

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

answer: Mixed or irregular scatterers consistent with lofted debris can support a tornado-debris interpretation when collocated with rotation; use corroborating evidence. (2 points).

13. (2 points) Calculate the surface temperature-dew-point spread.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

answer: 5 C (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , C.

14. (2 points) Identify one reason that 65 dBZ reflectivity alone does not prove a tornado.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

answer: High reflectivity can indicate large or numerous precipitation targets, including hail; it does not directly establish a rotating ground-contact vortex. (2 points).

15. (2 points) A storm moves 30 km east in 30 minutes. Find mean speed.

Storm-track table:
Time | Eastward distance
0 min | 0 km
30 min | 30 km

answer: 60 km/h (2 points).

Numerical target 60; absolute tolerance 0.12. Accepted units: , km/h.

16. (2 points) A rain gauge records 40 mm in 0.5 hours. Calculate mean rain rate.

Gauge table:
Elapsed time | Accumulated rain
0 h | 0 mm
0.5 h | 40 mm

answer: 80 mm/h (2 points).

Numerical target 80; absolute tolerance 0.16. Accepted units: , mm/h.

17. (2 points) A satellite IR time series shows cloud-top brightness temperature falling from 240 K to 220 K. Give a physically plausible interpretation and a limitation.

IR table:
Frame 1 | 240 K
Frame 2 | 220 K

answer: Colder cloud tops can be consistent with cloud growth to higher levels, but retrieval, cloud type and environmental temperature profile affect interpretation. (2 points).

18. (2 points) Station wind shifts form a coherent boundary while pressure rises behind it. Describe a candidate thunderstorm-related mechanism.

Station table:
Ahead of boundary | Original wind, lower pressure
Behind boundary | Shifted wind, rising pressure

answer: A thunderstorm cold pool/outflow boundary is consistent with the observations; compare radar, temperature changes and timing rather than identifying it from one variable alone. (2 points).

19. (2 points) Radar detects strong winds perpendicular to its beam but a small radial velocity. Explain why the wind need not be weak.

Radar geometry table:
Wind direction | Perpendicular to beam
Radial velocity | Small
Independent wind observation | Strong

answer: Doppler radial velocity is only the component along the beam; perpendicular motion contributes little to that component. (2 points).

20. (2 points) A short intense rain period falls on an already saturated basin. Explain why rainfall amount alone does not establish flood severity.

Basin table:
Antecedent soil | Saturated
Rain | Short, intense burst
Catchment drainage/routing | Not supplied

answer: Antecedent moisture, infiltration, drainage, catchment area and routing affect runoff and flood response; evaluate rainfall intensity and basin conditions together. (2 points).

21. (2 points) Use Figure F1. Read the velocity difference between the adjacent west and east cells.

Same small storm area: collocated measurements

Region	Reflectivity	Radial velocity	Correlation
West cell	65 dBZ	-30 m/s	0.70
East cell	60 dBZ	+35 m/s	0.72
Background	25 dBZ	+5 m/s	0.98

F1. Synthetic dual-polarization radar panels (original synthetic teaching figure)

answer: 65 m/s (2 points).

Numerical target 65; absolute tolerance 0.195. Accepted units: , m/s.

22. (2 points) Use Figure F1. Explain why the juxtaposed velocity signs support investigating rotation rather than concluding that both cells are calm.

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

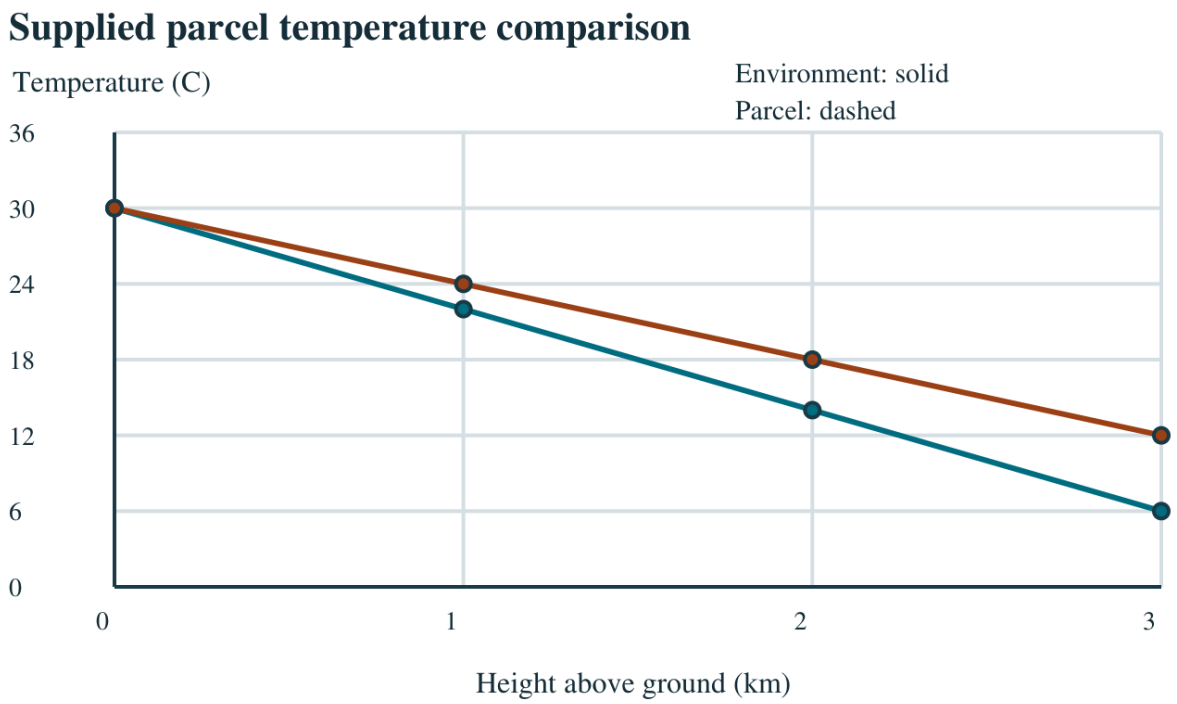
answer: The adjacent approaching and receding velocities form a couplet consistent with rotation; geometry, resolution and other evidence are needed for confirmation. (2 points).

23. (2 points) Use Figure F1. What extra interpretation does low correlation in the same area support, and what does it not prove by itself?

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

answer: It can support mixed/irregular scatterers such as lofted debris when collocated with rotation, but cannot independently prove a ground-contact tornado without corroborating evidence. (2 points).

24. (2 points) Use Figure F2. Read parcel-minus-environment temperature at 2 km.



F2. Parcel and environment profiles (original synthetic teaching figure)

answer: 4 C (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: , C.

25. (2 points) Use Figure F2. Under the stated simplification, is the parcel positively or negatively buoyant at 2 km? Explain.

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

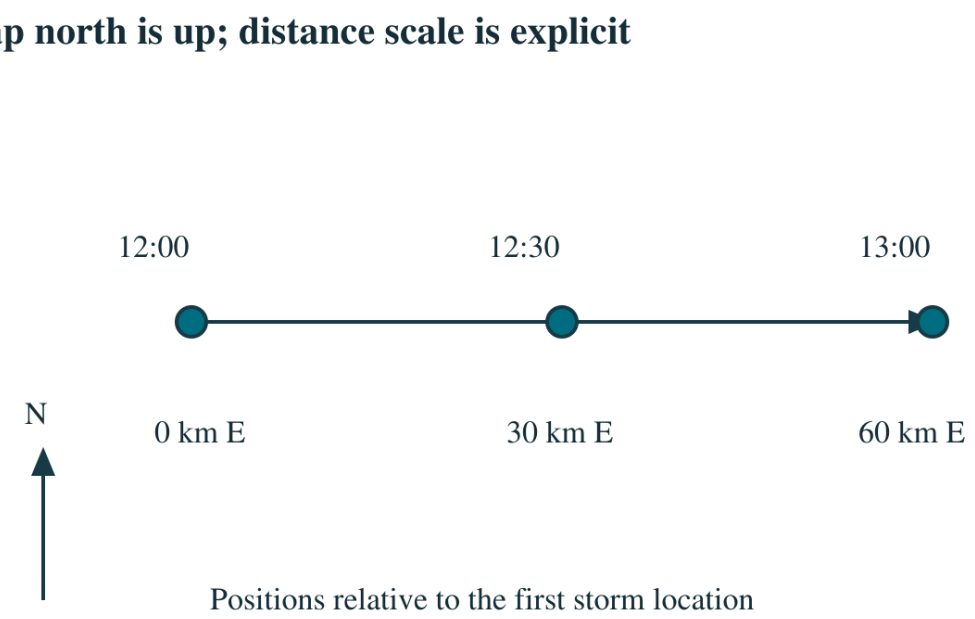
answer: Positively buoyant: the parcel is warmer and hence less dense than its environment at matched pressure under the stated simplification. (2 points).

26. (2 points) Use Figure F2. Why cannot this profile alone establish tornado occurrence?

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

answer: It does not supply all required storm and wind information, including shear/rotation, initiation and ground-level evidence; instability alone does not establish a tornado. (2 points).

27. (2 points) Use Figure F3. Calculate mean storm speed from 12:00 to 13:00.



F3. Storm-track map (original synthetic teaching figure)

answer: 60 km/h (2 points).

Numerical target 60; absolute tolerance 0.18. Accepted units: , km/h.

28. (2 points) Use Figure F3. Give the direction of the plotted motion.

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

answer: East. (2 points).

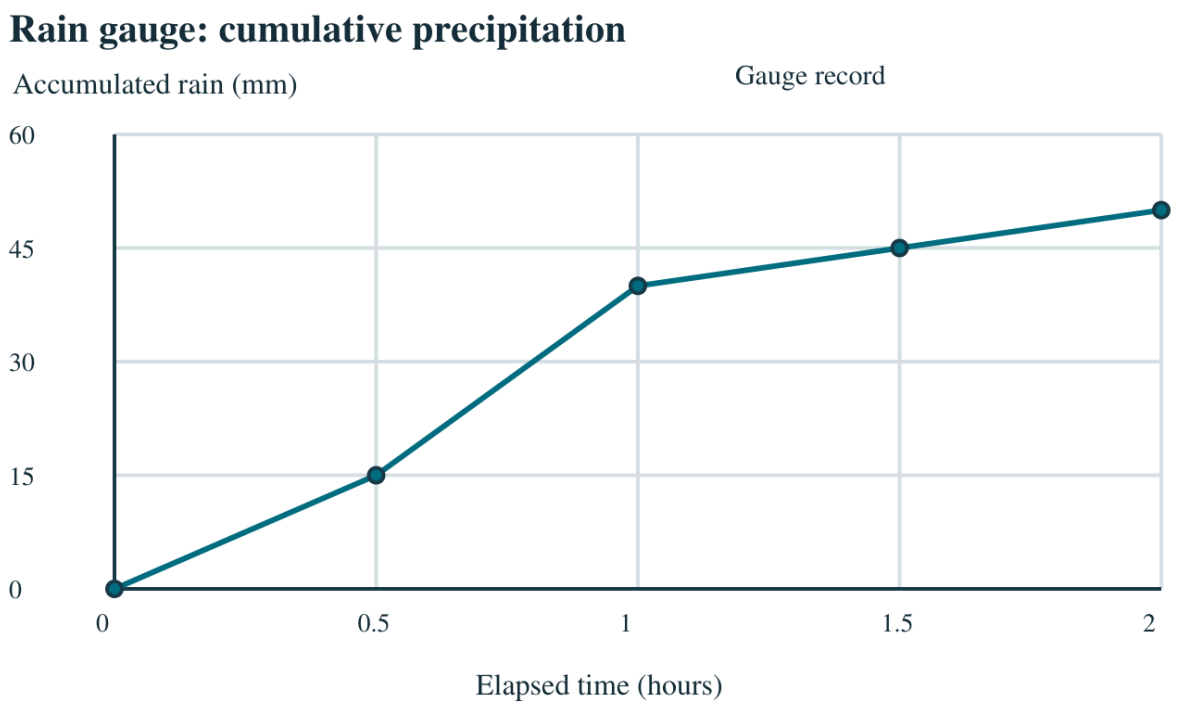
29. (2 points) Use Figure F3. If the motion continues unchanged, how many minutes after 13:00 would a point 30 km farther east be reached?

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

answer: 30 minutes (2 points).

Numerical target 30; absolute tolerance 0.09. Accepted units: , minutes.

30. (2 points) Use Figure F4. Calculate the mean rain rate during the 0.5-to-1.0 h interval.



F4. Accumulated rainfall curve (original synthetic teaching figure)

answer: 50 mm/h (2 points).

Numerical target 50; absolute tolerance 0.15. Accepted units: , mm/h.

31. (2 points) Use Figure F4. Identify the interval with the steepest accumulation and explain what the slope measures.

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.5 to1.0h; slope is rainfall rate, and this interval has the largest increase per unit time. (2 points).

32. (2 points) Use Figure F4. For an already saturated basin, explain why the steep interval deserves attention even if a later interval has a higher cumulative total.

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: Short intense rainfall can create rapid runoff when infiltration capacity/storage is limited. Cumulative total and current intensity describe different aspects of flood response. (2 points).